PolicyProbe – Student Privacy Forensic Auditor

Team: CODEXGLADIATORS

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PROJECT IDENTITY

Meet the Team & the Mission

Team & Project

Team Name: CODEX GLADIATORS

Project Title: PolicyProbe – Student Privacy Forensic Auditor

Problem Statement: PS-20 | Cybersecurity: Student Data Privacy Audit Tool

Mission

Deconstructing the "**black box**" of educational software to protect student identity and institutional integrity.

Why This Matters Now

Recent controversies like Proctorio and Chegg revealed how educational platforms allegedly used student data beyond academic purposes—a pattern PolicyProbe is designed to detect and prevent.

CRITERION 4 – PROBLEM UNDERSTANDING & RELEVANCE

The Indian EdTech Crisis

8,487
Attacks Per Week

India's education sectorfaces an average of 8,487 cyberattacks per organization weekly,makingit the highest targeted sectorinthe country.

3.9 Lakh+
Credentials Leaked

Recentthreatintelligence recorded over393,500instances ofIndian institutionaldatabreaches containingcredentialsin plain text over just 10 months.

23.6M+
Records Exposed

Weak security in Indian EdTech platforms recently exposed 22million student accounts and 1.6 million teacher records to threat actors.

The Forensic Ghosting Problem

Even after students delete an app, the trackers identified by PolicyProbe may have already mirrored their data across third-party ad networks, creating a permanent, un-erasable digital footprint. This is the core forensic challenge PolicyProbe addresses.

Relevance: Students are the least protected yet most targeted population for long-term identity theft.

CRITERION 1 – INNOVATION & IDEA QUALITY

What Makes PolicyProbe Different

Beyond Metadata

PolicyProbe isn't just a scanner; it's an **automated compliance auditor powered by a Heuristic Risk Engine**.

Permission-to-Category Mismatch Detection

Flags **"unreasonable access"** patterns—e.g., a simple Calculator app requesting **READ_SMS** or **ACCESS_FINE_LOCATION**. This reveals how malicious apps hide in plain sight by masquerading as benign tools.

The "Black-Box" Solver

Bridges technical network traffic and legal privacy policies, translating complex JSON/SSL data into plain-language risks using **LLM-driven entity extraction** to find hidden "data-sharing" clauses that standard static scanners miss.

CRITERION 2 – TECHNICAL APPROACH

Architecture Overview

Data Flow Pipeline

1

Input Layer
App Store URL

2

Static Scraper
Extracts permissions

3

Exodus Privacy Database
400+ ad-networks

4

Gemini 1.5 Flash NLP Auditor
Analyzes policy text

5

Risk Score Dashboard
0–100

Processing Layer Details

- **Static Scraper:** Extracts API permissions and data safety labels.
- **Logic Engine:** Cross-references developer domains against the **Exodus Privacy Database** (tracking 400+ ad-networks).
- **NLP Auditor:** Feeds scraped policy text into **Gemini 1.5 Flash** to identify COPPA/GDPR violations.

Output

Generates a real-time **Risk Score (0–100)** for every audited application or portal.

CRITERION 2 – TECHNICAL FEASIBILITY

Built to Ship Fast, Built to Scale

Fast Development

Built on **Python 3.12** and **Streamlit**. Streamlit allows for a professional dashboard UI in hours, not weeks.

Low Overhead

API-first design means no heavy infrastructure. It can audit a single classroom app or an entire University district portal from a single dashboard.

Integration

Can be easily converted into a **Chrome Extension** for administrators to audit platforms "on-the-fly."

CRITERION 3 – PRESENTATION & DESIGN

The "Privacy Nutrition Label"

Design Philosophy: Clarity over Complexity.

The Visual Report

Instead of raw logs, PolicyProbe outputs a **Privacy Nutrition Label**:

- **Trackers Found:** (e.g., 14 Analytics, 3 Advertising).
- **Data Selling:** High Alert (if found in policy).
- **Encryption Status:** Verified.

Accessibility

Designed for **non-technical administrators** to make informed procurement decisions.

CRITERION 5 – IMPACT & MARKET FIT

Who Benefits & Why It Matters

Institutional Impact

Empowers School IT leads to **"Red Flag"** apps before they reach the classroom.

User Empowerment

Students can **"Probe"** their own university portals to see where their login data is being mirrored.

Market Fit

With India's DPDP Act enforcing strict compliance, the domestic cybersecurity market is surging toward **₹30,000 Crores by 2027**. Institutions are desperate for automated, low-cost auditing solutions like PolicyProbe to avoid massive regulatory penalties.

CRITERION 5 – SCALABILITY & EXPANSION

The Road Ahead

Horizontal Scaling

Move from Mobile Apps to Browser-based LMS portals (Canvas, Moodle, Blackboard).

AI Red Teaming

Stress-test educational AI chatbots (**tutoring platforms, LLM-based assistants**) to ensure they aren't leaking student **PII** via **prompt injection**. This positions PolicyProbe as a specialized security tool for the AI-driven education ecosystem.

Global Reach

Move from Mobile Apps to Browser-based institutional LMS portals (e.g., TCS iON, CollPoll, Fedena, and Camu)